

AE460B Spacecraft Design

TM-01: Thermal Control Subsystem Design

Manifest Destiny

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This technical memorandum presents the preliminary design and first-order thermal analysis of the spacecraft Thermal Control subsystem developed for the Solar Gravitational Lens mission. We carefully define the spacecraft's thermal environment and outline the worst-case hot and cold scenarios it will encounter near the Sun and in deep space up to 900 AU away. A steady-state, lumped-parameter thermal model is used in our analysis to estimate the maximum and minimum component temperatures and also to determine radiator sizing and heater power requirements. Based on our computed results, we used a combination of passive and active thermal control strategies to maintain all components within an allowable temperature range while minimizing power consumption or system complexity.

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I. Thermal Environment Definition

A. External Environment

Solar Gravitational Lens (SGL) missions are unique in that it requires the spacecraft to travel further into deep space than we have ever gone previously. Because of this need for distance, we gather most of the exit velocity from the solar system through a sun gravity assist maneuver, placing us extremely close to the surface of the sun. In addition, traveling over 900 AU means spending an extended period experiencing deep space temperatures.

Because of these two extreme conditions, the spacecraft will experience a massive range of temperatures from its external environment. The Parker Solar Probe makes for a good reference point. Passing 0.042 AU from the surface of the sun, it underwent a max temperature of over 1600 K. With an estimated planned distance of 0.14 AU, using the airless equilibrium temperature equation below, we can expect the temperature to reach no more than 1000 K.

$$T \propto \frac{1}{\sqrt{r}} \quad (1)$$

The temperatures in deep space have been well known for years now. With space being nearly empty that far out, the sources of heat from stars are virtually nonexistent. Instead, the primary source of heat comes from Cosmic Microwave Background (CMB) radiation. Research has found the average temperature of CMB radiation, estimated to be 2.72548 ± 0.00057 K [1].

B. Internal Heat Dissipation

Internally, most of the systems on the spacecraft provide little to no heat. In order to bring the On-Board Computer (OBC) and many other components that cannot function near absolute zero to an appropriate temperature, we rely on our power system. The Multi-Hundred-Watt Radioisotope Thermoelectric Generator (MHW-RTG) was selected to provide 157 Watts to the spacecraft. In addition to the power it supplies, it would run with a skin temperature of 200°C . In order to heat the specific parts of the craft, we will utilize a tiling system to split the radiator into parts. These are capable of each producing 1 Watt, and have a skin temperature of 20°C [2]. With these in place, we can heat the spacecraft to hold a constant temperature of -100°C and keep the OBC running within allowed temperatures.

C. Worst-Case Bounding Scenarios

The worst-case bounds of the spacecraft lie in the events of failure from either the flight dynamics during the sun transfer, the heat shielding, or the radiator, which is the most heavily relied upon source of heat. During the close sun approach, if the flight passes closer than expected by just a few hundred kilometers, the rapid jump in temperature may exceed operational limitations. Such an event would likely fry electrical components and melt the solar sails. Without solar sails, the mission would never be completed due to the impossible distance required in the time frame. If the radiator runs less efficiently than experimentation predicts, or if the heat fails to distribute evenly to the desired locations, specific components may fail due to the cold climate in deep space. The fix for this is to have as much redundant heating in the system as possible.

II. Thermal Modeling

A. Modeling Assumptions

The thermal behavior of the SGL spacecraft is modeled using a first-order steady-state, lumped-parameter approach. This analysis assumes thermal equilibrium for each bounding case in the sense that the net heat rate to the spacecraft is zero. Because the mission includes both a solar perihelion maneuver and an extended deep-space cruise, thermal environments are treated on a phase-by-phase basis. In all phases beyond the atmosphere, heat transfer is assumed to be dominated by radiation. Convection is assumed to be negligible for our preliminary model. The external heating of our spacecraft is modeled primarily as solar radiation, and the solar irradiance is scaled using an inverse-square relationship with heliocentric distance.

The hot-case condition is defined by the solar Oberth maneuver at a minimum perihelion distance of 0.046 AU. The cold-case condition is defined by the cruise region, where solar input from the Sun becomes small, and survival depends on insulation and internal heat sources. The exterior surfaces that do not have radiators will be covered with multilayer insulation (MLI) to reduce radiative heat loss, and the surfaces with radiators will radiate out into deep space.

The thermo-optical properties of solar absorptance (α) and infrared emittance (ϵ) will be treated as constant for the baseline analysis. Potential α/ϵ degradation over the mission lifetime is treated as a design risk. When the spacecraft is at perihelion, it will be protected from direct solar exposure by the heat shield geometry and spacecraft pointing in a way that only a small fraction of incident flux reaches the bus. During planetary flybys, the spacecraft will remain distant from atmospheric flight. We will treat planetary infrared and albedo heating as second-order effects relative to the solar Oberth maneuver for our preliminary model.

B. Governing Equations

A first-order steady-state energy balance is used to model the spacecraft thermal behavior for each bounding case. Under the steady-state assumption, the net heat rate to the spacecraft is zero. For the radiation-dominated environment considered in this study, the primary heat input terms consist of absorbed solar radiation and internal heat dissipation, and the dominant heat rejection mechanism is thermal radiation to deep space. The energy balance for the lumped spacecraft node is therefore expressed as

$$\dot{Q}_{\text{solar}} + \dot{Q}_{\text{int}} + \dot{Q}_{\text{heater}} = \dot{Q}_{\text{rad}}. \quad (2)$$

The absorbed solar heat rate is modeled as

$$\dot{Q}_{\text{solar}} = \alpha S(r) A_{\text{proj}} \eta_{\text{shield}}, \quad (3)$$

where α is the effective solar absorptance of the exposed surface, A_{proj} is the projected area normal to the Sun, and η_{shield} represents an effective attenuation factor that accounts for spacecraft pointing and heat-shield protection during the solar perihelion phase. The solar irradiance $S(r)$ is scaled with heliocentric distance r according to the inverse-square law,

$$S(r) = S_0 \left(\frac{1 \text{ AU}}{r} \right)^2, \quad (4)$$

where $S_0 = 1361 \text{ W/m}^2$ is the solar constant at 1 AU. Internal heat dissipation, $\dot{Q}_{\text{int}}$, represents electrical power dissipated by onboard avionics and subsystems and is specified separately for the hot-case and cold-case operating modes. In cold-case conditions, an additional heater power term, $\dot{Q}_{\text{heater}}$, may be included to maintain component temperatures above survival limits.

Radiative heat rejection from the spacecraft is modeled using the Stefan–Boltzmann relation,

$$\dot{Q}_{\text{rad}} = \epsilon \sigma A_{\text{rad}} \left(T^4 - T_{\text{space}}^4 \right), \quad (5)$$

where ϵ is the effective infrared emittance of the radiating surface, σ is the Stefan–Boltzmann constant, A_{rad} is the effective radiator area, T is the spacecraft equilibrium temperature, and T_{space} is the effective background temperature of deep space. For all analyzed cases, T_{space} is assumed to be sufficiently small relative to T and is neglected. When the governing equations are solved, it yields the steady-state equilibrium temperature for the hot and cold case conditions.

III. Hot-Case Analysis

A. Hot-Case Assumptions

In the hot case, we will be splitting the spacecraft into two bodies: the sail and the hub. This assumption is made because the sail is quite large and the connection to the hub is relatively small. In other words, the sail and the hub likely do not exchange heat quickly enough to obtain equilibrium. For the sail, we will first assume a Kapton structure with an aluminum coating on the sunny side and a chromium coating on the dark side. The aluminum loses structural stability first at 290 °C.[3] Additionally, we will assume that the sun only hits the aluminum side, where α_{Al} is 0.09 and $\epsilon_{Al} = 0.026$. The dark side has ϵ_{Cr} of 0.24.[4] We will also run a case for a general solar dive and determine the necessary characteristics for the sail to survive at any radius.

For the bus, we assume the craft is mainly aluminum with the same thermal characteristics. However, we assume that the sun mostly hits a silica heat shield with α_s of 0.02 and ϵ_s of 0.7 with an area a fifth of the total spacecraft area.[5] The rest of the area is the aluminum and, if necessary, radiator. We also assume that the camera limits the max

equilibrium temperature of the craft to 50 °C.[6] Lastly, the sink temperature is assumed to be small compared to the temperature of the spacecraft, and in the analysis, we assume the bulk body to be at the same temperature throughout.

In all of these cases, we will assume a 10% thermal margin scenario in terms of °C, so for the camera it would be 45°C and the aluminum 271 °C.

B. Hot-Case Results

For the sail, we get that the closest the spacecraft can get with current materials is 0.311 AU, which is not close enough to achieve the goals of the mission. Using the following simplification:

$$S_{max} = \frac{(\epsilon_{Al} + \epsilon_{Cr})}{\alpha_{Al}} \sigma T_{max}^4 = 2 \left(\frac{\epsilon}{\alpha} \right)_{avg} \sigma T_{max}^4 \quad (6)$$

And plugging in the solar irradiance for 0.14 AU, we get that the average $\frac{\epsilon}{\alpha}$ ratio must be 7.23, which is much higher compared to the current 2.96. This indicates that there is a necessary improvement in the sail materials before this mission can take off.

For the bus, we can calculate the temperature with just a heat shield and without a radiator at 0.14 AU to be 140 °C, which is higher than the max. When including a radiator with emissivity 0.8, we get that the required area for the worst-case scenario is 37% of the total surface area of the spacecraft.[7]

IV. Cold-Case Analysis

A. Cold-Case Assumptions

In the cold case, to assume worst case scenario, we take that the bottom limit of the spacecraft is to be the most temperature-sensitive component: the camera. This value turns out to be about -10 °C or 263 K. For a thermal margin of 10%, we get 16.3 °C. In other words, we assume the spacecraft and the camera have about the same temperature. This also plays into the heating, where we require enough energy to heat the entire spacecraft to 289.3 K. We also assume that the sink temperature from the background radiation is too small to be considered. Lastly, we assume the emissivity of the spacecraft to be ≈ 0.03 .

B. Cold-Case Results

Using those numbers, we get that the internal power required to heat the spacecraft is a measly 12.4 watts. Even if we were to take into consideration the half-life of Pu-238, for a 25-year mission, the power at launch would have to be 18.6 watts. What might be more of an issue in deep space is perhaps trying to remove any extra energy from the spacecraft that may overheat any component; however, this is unlikely, and any excess energy can be stored in flywheels or batteries. A more sure-fire analysis requires a computer model of the components.

Below is a table of the summarized results.

Parameter	Value
Maximum Predicted Bulk Temperature	45 °C
Minimum Predicted Bulk Temperature	16 °C
Radiator area	$0.37 A_{craft}$
Heater power (after 25 years)	12.4 W
Thermal margin	10%

Table 1 Preliminary Thermal Parameters of the Spacecraft

V. Thermal Control Strategy Selection

A. Passive Thermal Control

Passive thermal control is used as the primary means of regulating the SGL spacecraft temperature to minimize our power consumption and also system complexity [8]. The main strategy will revolve around tailoring its surface properties, thermal insulation, and heat rejection into space to manage the wide range of thermal environments encountered throughout the mission. All of the exterior surfaces of the spacecraft bus that do not radiate will be covered with multilayer insulation (MLI) to reduce the loss of heat during our deep-space cruise and to limit the absorbed solar energy during our high-flux mission phases. The surface coatings that we choose, such as aluminized CP1, will be based on their thermo-optical properties. Specifically, we want to make sure that we achieve a low solar absorptance-to-infrared emittance ratio (α/ϵ) to minimize equilibrium temperatures under solar loading. Radiator surfaces will be utilized to reject internally dissipated heat by thermal radiation into deep space. The radiators will be thermally isolated from any non-radiating surfaces to ensure that heat rejection is efficient. The passive architecture is designed to maintain acceptable temperatures over the majority of the mission without having to rely on continuous active thermal control.

B. Active Thermal Control

Active thermal control will be implemented to supplement the passive thermal architecture and to ensure the survivability of the SGL spacecraft when it is in deep space up to 900 AU for the cold-case conditions. The primary active thermal control elements considered in this design will be electrical resistance heaters that will be integrated within the spacecraft bus. The heaters will primarily operate during cold-case mission phases. This is especially the case for the extended deep space cruise, where solar heating is minimal and internal power dissipation is reduced. Heater power requirements are determined from the steady-state cold-case energy balance calculations and are sized to maintain our component temperatures above minimal survival limits. The heaters will likely be thermostatically controlled, where they will only activate once temperatures fall below a predefined threshold. This will allow us to minimize power consumption while also providing robust temperature regulation during low-flux mission phases. We will not be making use of any cooling systems for the hot-case scenario since thermal control will be achieved through the heat shield and geometry of the spacecraft itself.

VI. Subsystem Interfaces and Risk Identification

A. Power Subsystem Interface

The thermal control subsystem interfaces closely with the spacecraft power subsystem since heater operation and thermal regulation are inherently coupled to available electrical power and mission operating modes [9]. The thermoelectric generator will distribute power to the thermal system, which shall be allocated via DC converters and mechanical switching systems, with shunt regulators used to control the voltage requirement. This will give it the ability to be in use only when needed.

B. Structures Subsystem Interface

The structures subsystem is the skeleton of the spacecraft and will be the subsystem most directly affected by temperature change. In the hot case, as the aluminum frame itself heats up, it will expand. In particular, longer beams will increase in length. This expansion will be addressed by having regularly spaced joints separating beams so that individual beam displacement is smaller. Furthermore, the joints and fasteners will have thermal flex tolerance and incorporate compliant mounts to allow for a small amount of bending.

In the cold case, the structure itself will be at significantly less risk. The only concern would be excessive thermal contraction of structural members, which could introduce internal stresses or cause slight misalignments between attached components. While aluminum generally performs well at low temperatures, repeated thermal cycling could lead to fatigue over time if not properly managed. These concerns will be addressed by the conduction coming from the heaters to keep the frame at a reasonable cold temperature.

C. Operations and Attitude Interface

The thermal risks involved in the operations and attitude interface are as follows. Due to our reaction wheels and star tracker coupling, our coupling will generate heat during maneuvers. This will heat our wheels and star tracker mounts, which can cause short-term attitude bias.

If our wheels freeze, they may not be able to spin, which will cause them to tumble. If frozen, this will cause severe internal bearing failure. These risks can be mitigated using heaters with tight thermostat readings and vibration monitoring vs temperature.

If our star tracker were to overheat, we would have no fine attitude knowledge and unclear camera imaging. If our star tracker fails due to freezing over, we will experience long reacquisition times, and we may miss out on critical imaging. These risks can be mitigated using dedicated radiators with thermal insulation for tight thermal control.

D. Risk Identification and Mitigation

Thermal Risk 1: During the solar Oberth maneuver at 0.14 AU, the spacecraft is exposed to extreme solar irradiance. Insufficient protection by the thermal protection system could result in excessive heat absorption by the spacecraft bus, exceeding component temperature limits. Mitigation: A carbon-carbon composite heat shield with a high-reflectivity ceramic coating is employed to attenuate incident solar flux during perihelion. The heat shield is jettisoned following perihelion to eliminate unnecessary thermal mass.

Thermal Risk 2: Beyond approximately 100 AU, external solar heating becomes negligible, and sustained radiative heat loss may reduce spacecraft component temperatures below survival limits during long-duration cruise. Mitigation: Thermal control in deep space is achieved primarily through passive heat retention using MLI and controlled radiative surfaces. Waste heat from the RTG serves as the dominant internal heat source. Radioisotope heater units provide localized heating for thermally sensitive components where required.

Thermal Risk 3: Limited electrical power availability, particularly late in the mission due to RTG degradation, may restrict the use of active thermal control systems. Mitigation: The thermal architecture is designed to be predominantly passive, minimizing reliance on electrical heaters. Where active control is required, RHUs are favored due to their independence from the spacecraft power budget.

VII. Conclusion

This technical memorandum presented the preliminary design and first-order thermal analysis of the Thermal Control Subsystem for the Solar Gravitational Lens mission. The spacecraft thermal environment was defined, and worst-case hot and cold scenarios were evaluated using a steady-state, lumped parameter thermal model. Our results from the hot and cold analyses demonstrate that a predominantly passive thermal control architecture that is supplemented by active heaters during deep-space cruise is capable of maintaining spacecraft temperatures within allowable limits across all of our analyzed mission phases. The selected thermal control strategy leverages the tailoring of our surface properties, multilayer insulation, dedicated radiative heat rejection, and minimal active heating to balance performance with system simplicity as well as power efficiency. While the present analysis provides a conservative and defensible baseline for subsystem sizing, future design phases should incorporate higher-fidelity thermal math models, transient analysis, and detailed geometric representations to refine the predictions of our temperatures and to assess our thermal margins when they are under off-nominal conditions.

Appendix

Figures

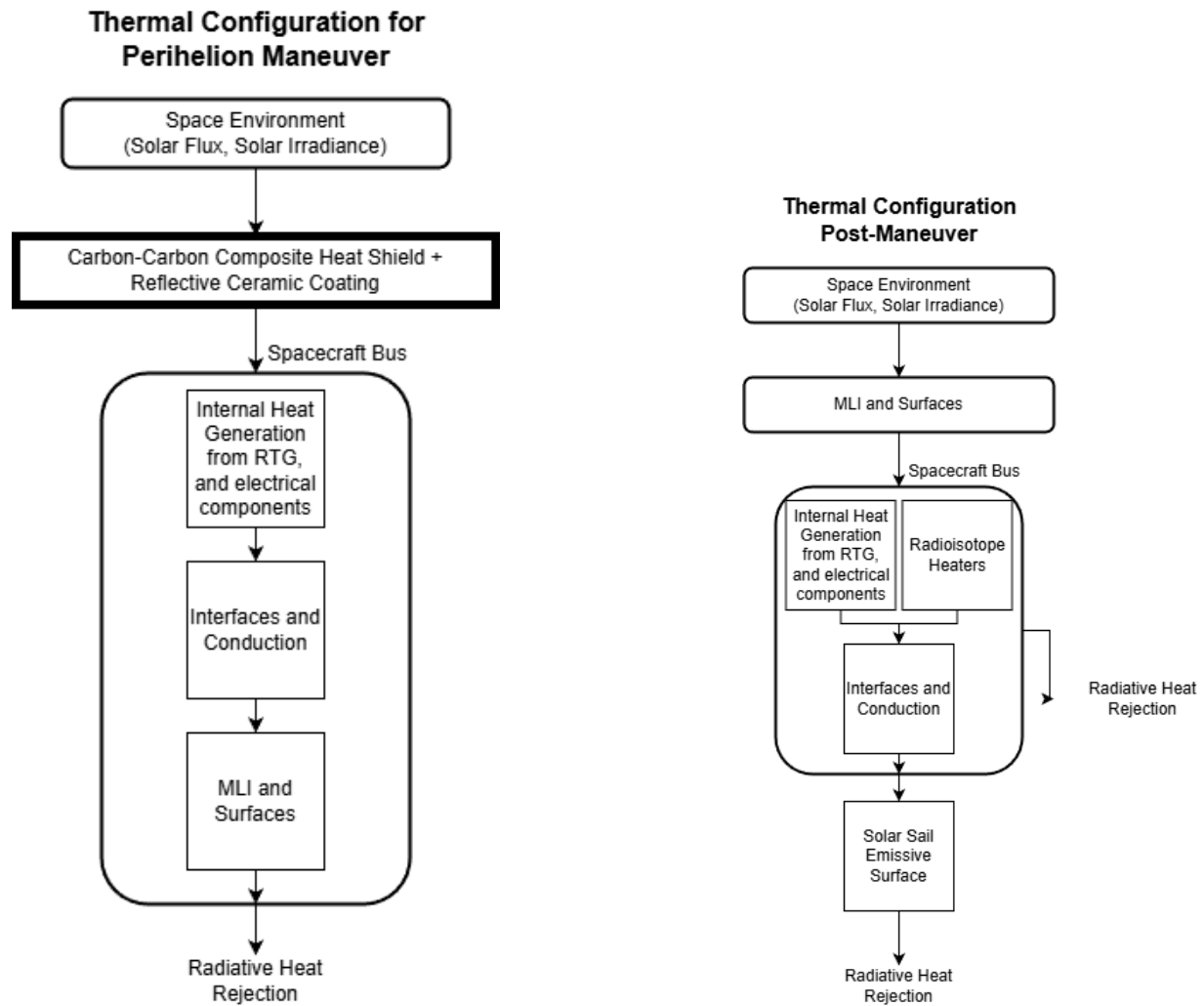


Fig. 1 Left: Thermal Block Diagram during Perihelion Maneuver. Right: Thermal Block Diagram Post-Maneuver.

Nomenclature

A_{craft}	=	Total external surface area of the spacecraft (bus)
A_{proj}	=	Projected area normal to incident solar radiation
A_{rad}	=	Effective radiator area
α	=	Solar absorptance of spacecraft surface
α_{Al}	=	Solar absorptance of aluminum coating
α_s	=	Solar absorptance of silica heat shield surface
AU	=	Astronomical Unit
CMB	=	Cosmic Microwave Background radiation
CP1	=	Colorless Polyimide-1 (spacecraft film material)
ε	=	Infrared emittance of spacecraft surface
ε_{Al}	=	Infrared emittance of aluminum coating
ε_{Cr}	=	Infrared emittance of chromium coating
ε_s	=	Infrared emittance of silica heat shield surface
$(\varepsilon/\alpha)_{\text{avg}}$	=	Average emittance-to-absorptance ratio
η_{shield}	=	Effective solar flux attenuation due to heat shield and pointing
MHW-RTG	=	Multi-Hundred-Watt Radioisotope Thermoelectric Generator
MLI	=	Multi-Layer Insulation
OBC	=	On-Board Computer
^{238}Pu	=	Plutonium-238 (radioisotope fuel)
$\dot{Q}_{\text{heater}}$	=	Heater power input
$\dot{Q}_{\text{int}}$	=	Internal heat dissipation rate
$\dot{Q}_{\text{rad}}$	=	Radiative heat rejection rate
$\dot{Q}_{\text{solar}}$	=	Absorbed solar heat rate
r	=	Heliocentric distance
RHU	=	Radioisotope Heater Unit
RTG	=	Radioisotope Thermoelectric Generator
$S(r)$	=	Solar irradiance at heliocentric distance r
S_0	=	Solar constant at 1 AU (1361 W/m^2)
S_{max}	=	Maximum allowable solar irradiance for a given material limit
SGL	=	Solar Gravitational Lens
σ	=	Stefan-Boltzmann constant ($5.670 \times 10^{-8} \text{ W/m}^2\text{K}^4$)
T	=	Spacecraft equilibrium temperature
T_{max}	=	Maximum allowable temperature (material/operational limit)
T_{space}	=	Effective deep-space background temperature

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